After reviewing this checklist, what will you do to become confident for all objectives?

4.7 Solve Equations with Fractions

Learning Objectives

By the end of this section, you will be able to:

- Determine whether a fraction is a solution of an equation
- Solve equations with fractions using the Addition, Subtraction, and Division Properties of Equality
- Solve equations using the Multiplication Property of Equality
- Translate sentences to equations and solve

BE PREPARED 4.17 Before you get started, take this readiness quiz. If you miss a problem, go back to the section listed and review the material.

Evaluate $x + 4$ when $x = -3$
If you missed this problem, review Example 3.23.

BE PREPARED 4.18 Solve: $2y - 3 = 9$.
If you missed this problem, review Example 3.61.

BE PREPARED 4.19 Solve: $y - 3 = -9$
If you missed this problem, review Example 4.28.

Determine Whether a Fraction is a Solution of an Equation

As we saw in Solve Equations with the Subtraction and Addition Properties of Equality and Solve Equations Using Integers; The Division Property of Equality, a solution of an equation is a value that makes a true statement when substituted for the variable in the equation. In those sections, we found whole number and integer solutions to equations. Now that we have worked with fractions, we are ready to find fraction solutions to equations.

The steps we take to determine whether a number is a solution to an equation are the same whether the solution is a whole number, an integer, or a fraction.

HOW TO

Determine whether a number is a solution to an equation.

Step 1. Substitute the number for the variable in the equation.
Step 2. Simplify the expressions on both sides of the equation.
Step 3. Determine whether the resulting equation is true. If it is true, the number is a solution. If it is not true, the number is not a solution.

EXAMPLE 4.95

Determine whether each of the following is a solution of $x - \frac{3}{10} = \frac{1}{2}$.

ⓐ $x = 1$ ⓑ $x = \frac{4}{5}$ ⓒ $x = -\frac{4}{5}$

Solution

ⓐ

$$x - \frac{3}{10} = \frac{1}{2}$$

Substitute 1 for x.	$1-\frac{3}{10}\stackrel{?}{=}\frac{1}{2}$
Change to fractions with a LCD of 10.	$\frac{10}{10}-\frac{3}{10}\stackrel{?}{=}\frac{5}{10}$
Subtract.	$\frac{7}{10}\neq\frac{5}{10}$

Since $x=1$ does not result in a true equation, 1 is not a solution to the equation.

ⓑ

	$x-\frac{3}{10}=\frac{1}{2}$
Substitute $\frac{4}{5}$ for x.	$\frac{4}{5}-\frac{3}{10}\stackrel{?}{=}\frac{1}{2}$
	$\frac{8}{10}-\frac{3}{10}\stackrel{?}{=}\frac{5}{10}$
Subtract.	$\frac{5}{10}=\frac{5}{10}$ ✓

Since $x=\frac{4}{5}$ results in a true equation, $\frac{4}{5}$ is a solution to the equation $x-\frac{3}{10}=\frac{1}{2}$.

ⓒ

	$x-\frac{3}{10}=\frac{1}{2}$
Substitute $-\frac{4}{5}$ for x.	$-\frac{4}{5}-\frac{3}{10}\stackrel{?}{=}\frac{1}{2}$
	$-\frac{8}{10}-\frac{3}{10}\stackrel{?}{=}\frac{5}{10}$
Subtract.	$-\frac{11}{10}\neq\frac{5}{10}$

Since $x=-\frac{4}{5}$ does not result in a true equation, $-\frac{4}{5}$ is not a solution to the equation.

TRY IT 4.189 Determine whether each number is a solution of the given equation.

$x-\frac{2}{3}=\frac{1}{6}$:

ⓐ $x=1$ ⓑ $x=\frac{5}{6}$ ⓒ $x=-\frac{5}{6}$

TRY IT 4.190 Determine whether each number is a solution of the given equation.

$y - \frac{1}{4} = \frac{3}{8}$:

ⓐ $y = 1$ ⓑ $y = -\frac{5}{8}$ ⓒ $y = \frac{5}{8}$

Solve Equations with Fractions using the Addition, Subtraction, and Division Properties of Equality

In Solve Equations with the Subtraction and Addition Properties of Equality and Solve Equations Using Integers; The Division Property of Equality, we solved equations using the Addition, Subtraction, and Division Properties of Equality. We will use these same properties to solve equations with fractions.

Addition, Subtraction, and Division Properties of Equality

For any numbers a, b, and c,

if $a = b$, then $a + c = b + c$.	Addition Property of Equality
if $a = b$, then $a - c = b - c$.	Subtraction Property of Equality
if $a = b$, then $\frac{a}{c} = \frac{b}{c}, c \neq 0$.	Division Property of Equality

Table 4.3

In other words, when you add or subtract the same quantity from both sides of an equation, or divide both sides by the same quantity, you still have equality.

EXAMPLE 4.96

Solve: $y + \frac{9}{16} = \frac{5}{16}$.

Solution

	$y + \frac{9}{16} = \frac{5}{16}$
Subtract $\frac{9}{16}$ from each side to undo the addition.	$y + \frac{9}{16} - \frac{9}{16} = \frac{5}{16} - \frac{9}{16}$
Simplify on each side of the equation.	$y + 0 = -\frac{4}{16}$
Simplify the fraction.	$y = -\frac{1}{4}$
Check:	$y + \frac{9}{16} = \frac{5}{16}$
Substitute $y = -\frac{1}{4}$.	$-\frac{1}{4} + \frac{9}{16} \stackrel{?}{=} \frac{5}{16}$

Rewrite as fractions with the LCD.	$-\frac{4}{16}+\frac{9}{16}\stackrel{?}{=}\frac{5}{16}$
Add.	$\frac{5}{16}=\frac{5}{16}$ ✓

Since $y=-\frac{1}{4}$ makes $y+\frac{9}{16}=\frac{5}{16}$ a true statement, we know we have found the solution to this equation.

TRY IT 4.191 Solve: $y+\frac{11}{12}=\frac{5}{12}$.

TRY IT 4.192 Solve: $y+\frac{8}{15}=\frac{4}{15}$.

We used the Subtraction Property of Equality in Example 4.96. Now we'll use the Addition Property of Equality.

EXAMPLE 4.97

Solve: $a-\frac{5}{9}=-\frac{8}{9}$.

Solution

	$a-\frac{5}{9}=-\frac{8}{9}$
Add $\frac{5}{9}$ from each side to undo the subtraction.	$a-\frac{5}{9}+\frac{5}{9}=-\frac{8}{9}+\frac{5}{9}$
Simplify on each side of the equation.	$a+0=-\frac{3}{9}$
Simplify the fraction.	$a=-\frac{1}{3}$
Check:	$a-\frac{5}{9}=-\frac{8}{9}$
Substitute $a=-\frac{1}{3}$.	$-\frac{1}{3}-\frac{5}{9}\stackrel{?}{=}-\frac{8}{9}$
Change to common denominator.	$-\frac{3}{9}-\frac{5}{9}\stackrel{?}{=}-\frac{8}{9}$
Subtract.	$-\frac{8}{9}=-\frac{8}{9}$ ✓

Since $a=-\frac{1}{3}$ makes the equation true, we know that $a=-\frac{1}{3}$ is the solution to the equation.

TRY IT 4.193 Solve: $a-\frac{3}{5}=-\frac{8}{5}$.

TRY IT 4.194 Solve: $n-\frac{3}{7}=-\frac{9}{7}$.

The next example may not seem to have a fraction, but let's see what happens when we solve it.

EXAMPLE 4.98

Solve: $10q = 44$.

Solution

		$10q = 44$
Divide both sides by 10 to undo the multiplication.		$\frac{10q}{10} = \frac{44}{10}$
Simplify.		$q = \frac{22}{5}$
Check:		
Substitute $q = \frac{22}{5}$ into the original equation.	$10(\frac{22}{5}) \stackrel{?}{=} 44$	
Simplify.	$\cancel{10}^{2}\left(\frac{22}{\cancel{5}}\right) \stackrel{?}{=} 44$	
Multiply.	$44 = 44$ ✓	

The solution to the equation was the fraction $\frac{22}{5}$. We leave it as an improper fraction.

TRY IT 4.195 Solve: $12u = -76$.

TRY IT 4.196 Solve: $8m = 92$.

Solve Equations with Fractions Using the Multiplication Property of Equality

Consider the equation $\frac{x}{4} = 3$. We want to know what number divided by 4 gives 3. So to "undo" the division, we will need to multiply by 4. The *Multiplication Property of Equality* will allow us to do this. This property says that if we start with two equal quantities and multiply both by the same number, the results are equal.

The Multiplication Property of Equality

For any numbers a, b, and c,

$$\text{if } a = b, \text{ then } ac = bc.$$

If you multiply both sides of an equation by the same quantity, you still have equality.

Let's use the Multiplication Property of Equality to solve the equation $\frac{x}{7} = -9$.

EXAMPLE 4.99

Solve: $\frac{x}{7} = -9$.

Solution

		$\frac{x}{7} = -9$
Use the Multiplication Property of Equality to multiply both sides by 7. This will isolate the variable.		$7 \cdot \frac{x}{7} = 7(-9)$
Multiply.		$\frac{7x}{7} = -63$
Simplify.		$x = -63$
Check. Substitute -63 for x for in the original equation.	$\frac{-63}{7} \stackrel{?}{=} -9$	
The equation is true.	$-9 = -9$ ✓	

TRY IT 4.197 Solve: $\frac{f}{5} = -25$.

TRY IT 4.198 Solve: $\frac{h}{9} = -27$.

EXAMPLE 4.100

Solve: $\frac{p}{-8} = -40$.

Solution

Here, p is divided by -8. We must multiply by -8 to isolate p.

		$\frac{p}{-8} = -40$
Multiply both sides by -8		$-8\left(\frac{p}{-8}\right) = -8(-40)$
Multiply.		$\frac{-8p}{-8} = 320$
Simplify.		$p = 320$
Check:		
Substitute $p = 320$.	$\frac{320}{-8} \stackrel{?}{=} -40$	
The equation is true.	$-40 = -40$ ✓	

TRY IT 4.199 Solve: $\frac{c}{-7} = -35$.

TRY IT 4.200 Solve: $\frac{x}{-11} = -12$.

Solve Equations with a Coefficient of -1

Look at the equation $-y = 15$. Does it look as if y is already isolated? But there is a negative sign in front of y, so it is not isolated.

There are three different ways to isolate the variable in this type of equation. We will show all three ways in Example 4.101.

EXAMPLE 4.101

Solve: $-y = 15$.

Solution

One way to solve the equation is to rewrite $-y$ as $-1y$, and then use the Division Property of Equality to isolate y.

	$-y = 15$
Rewrite $-y$ as $-1y$.	$-1y = 15$
Divide both sides by –1.	$\frac{-1y}{-1} = \frac{15}{-1}$
Simplify each side.	$y = -15$

Another way to solve this equation is to multiply both sides of the equation by -1.

	$-y = 15$
Multiply both sides by –1.	$-1(-y) = -1(15)$
Simplify each side.	$y = -15$

The third way to solve the equation is to read $-y$ as "the opposite of y." What number has 15 as its opposite? The opposite of 15 is -15. So $y = -15$.

For all three methods, we isolated y is isolated and solved the equation.

Check:

	$-y = 15$
Substitute $y = -15$.	$-(-15) \stackrel{?}{=} (15)$
Simplify. The equation is true.	$15 = 15$ ✓

TRY IT 4.201 Solve: $-y = 48$.

TRY IT 4.202 Solve: $-c = -23$.

Solve Equations with a Fraction Coefficient

When we have an equation with a fraction coefficient we can use the Multiplication Property of Equality to make the

coefficient equal to 1.

For example, in the equation:

$$\frac{3}{4}x = 24$$

The coefficient of x is $\frac{3}{4}$. To solve for x, we need its coefficient to be 1. Since the product of a number and its reciprocal is 1, our strategy here will be to isolate x by multiplying by the reciprocal of $\frac{3}{4}$. We will do this in Example 4.102.

EXAMPLE 4.102

Solve: $\frac{3}{4}x = 24$.

Solution

	$\frac{3}{4}x = 24$
Multiply both sides by the reciprocal of the coefficient.	$\frac{4}{3} \cdot \frac{3}{4}x = \frac{4}{3} \cdot 24$
Simplify.	$1x = \frac{4}{3} \cdot \frac{24}{1}$
Multiply.	$x = 32$
Check:	$\frac{3}{4}x = 24$
Substitute $x = 32$.	$\frac{3}{4} \cdot 32 \stackrel{?}{=} 24$
Rewrite 32 as a fraction.	$\frac{3}{4} \cdot \frac{32}{1} \stackrel{?}{=} 24$
Multiply. The equation is true.	$24 = 24$ ✓

Notice that in the equation $\frac{3}{4}x = 24$, we could have divided both sides by $\frac{3}{4}$ to get x by itself. Dividing is the same as multiplying by the reciprocal, so we would get the same result. But most people agree that multiplying by the reciprocal is easier.

TRY IT 4.203 Solve: $\frac{2}{5}n = 14$.

TRY IT 4.204 Solve: $\frac{5}{6}y = 15$.

EXAMPLE 4.103

Solve: $-\frac{3}{8}w = 72$.

Solution

The coefficient is a negative fraction. Remember that a number and its reciprocal have the same sign, so the reciprocal of the coefficient must also be negative.

	$-\frac{3}{8}w = 72$
Multiply both sides by the reciprocal of $-\frac{3}{8}$.	$-\frac{8}{3}\left(-\frac{3}{8}w\right) = \left(-\frac{8}{3}\right)72$
Simplify; reciprocals multiply to one.	$1w = -\frac{8}{3} \cdot \frac{72}{1}$
Multiply.	$w = -192$
Check:	$-\frac{3}{8}w = 72$
Let $w = -192$.	$-\frac{3}{8}(-192) \stackrel{?}{=} 72$
Multiply. It checks.	$72 = 72$ ✓

TRY IT 4.205 Solve: $-\frac{4}{7}a = 52$.

TRY IT 4.206 Solve: $-\frac{7}{9}w = 84$.

Translate Sentences to Equations and Solve

Now we have covered all four properties of equality—subtraction, addition, division, and multiplication. We'll list them all together here for easy reference.

Subtraction Property of Equality: For any real numbers a, b, and c, if $a = b$, then $a - c = b - c$.	**Addition Property of Equality:** For any real numbers a, b, and c, if $a = b$, then $a + c = b + c$.
Division Property of Equality: For any numbers a, b, and c, where $c \neq 0$ if $a = b$, then $\frac{a}{c} = \frac{b}{c}$	**Multiplication Property of Equality:** For any real numbers a, b, and c if $a = b$, then $ac = bc$

When you add, subtract, multiply or divide the same quantity from both sides of an equation, you still have equality.

In the next few examples, we'll translate sentences into equations and then solve the equations. It might be helpful to review the translation table in Evaluate, Simplify, and Translate Expressions.

EXAMPLE 4.104

Translate and solve: n divided by 6 is -24.

Solution

Translate.	$\underbrace{\frac{n}{6}}_{n \text{ divided by } 6} \underbrace{=}_{\text{is}} \underbrace{-24}_{24}$
Multiply both sides by 6.	$6 \cdot \frac{n}{6} = 6(-24)$
Simplify.	$n = -144$

Check:	Is -144 divided by 6 equal to -24?	
Translate.	$\frac{-144}{6} \stackrel{?}{=} -24$	
Simplify. It checks.	$-24 = -24$ ✓	

TRY IT 4.207 Translate and solve: n divided by 7 is equal to -21.

TRY IT 4.208 Translate and solve: n divided by 8 is equal to -56.

EXAMPLE 4.105

Translate and solve: The quotient of q and -5 is 70.

Solution

Translate.	$\underbrace{\frac{q}{-5}}_{\text{The quotient of } q \text{ and } -5} \underbrace{=}_{\text{is}} \underbrace{70}_{70}$
Multiply both sides by -5.	$-5\left(\frac{q}{-5}\right) = -5(70)$
Simplify.	$q = -350$

Check:	Is the quotient of -350 and -5 equal to 70?	
Translate.	$\frac{-350}{-5} \stackrel{?}{=} 70$	
Simplify. It checks.	$70 = 70$ ✓	

TRY IT 4.209 Translate and solve: The quotient of q and -8 is 72.

TRY IT 4.210 Translate and solve: The quotient of p and -9 is 81.

EXAMPLE 4.106

Translate and solve: Two-thirds of f is 18.

Solution

Translate.	$\underbrace{\frac{2}{3}f}_{\text{Two-thirds of } f} \quad \underbrace{=}_{\text{is}} \quad \underbrace{18}_{18}$
Multiply both sides by $\frac{3}{2}$.	$\frac{3}{2} \cdot \frac{2}{3}f = \frac{3}{2} \cdot 18$
Simplify.	$f = 27$

Check:	Is two-thirds of 27 equal to 18?
Translate.	$\frac{2}{3}(27) \stackrel{?}{=} 18$
Simplify. It checks.	$18 = 18$ ✓

TRY IT 4.211 Translate and solve: Two-fifths of f is 16.

TRY IT 4.212 Translate and solve: Three-fourths of f is 21.

EXAMPLE 4.107

Translate and solve: The quotient of m and $\frac{5}{6}$ is $\frac{3}{4}$.

Solution

	The quotient of m and $\frac{5}{6}$ is $\frac{3}{4}$.
Translate.	$\frac{m}{\frac{5}{6}} = \frac{3}{4}$
Multiply both sides by $\frac{5}{6}$ to isolate m.	$\frac{5}{6}\left(\frac{m}{\frac{5}{6}}\right) = \frac{5}{6}\left(\frac{3}{4}\right)$
Simplify.	$m = \frac{5 \cdot 3}{6 \cdot 4}$
Remove common factors and multiply.	$m = \frac{5}{8}$

Check:		
Is the quotient of $\frac{5}{8}$ and $\frac{5}{6}$ equal to $\frac{3}{4}$?	$\frac{\frac{5}{8}}{\frac{5}{6}} \stackrel{?}{=} \frac{3}{4}$	

Rewrite as division.	$\frac{5}{8} \div \frac{5}{6} \stackrel{?}{=} \frac{3}{4}$	
Multiply the first fraction by the reciprocal of the second.	$\frac{5}{8} \cdot \frac{6}{5} \stackrel{?}{=} \frac{3}{4}$	
Simplify.	$\frac{3}{4} = \frac{3}{4}$✓	

Our solution checks.

TRY IT 4.213 Translate and solve. The quotient of n and $\frac{2}{3}$ is $\frac{5}{12}$.

TRY IT 4.214 Translate and solve The quotient of c and $\frac{3}{8}$ is $\frac{4}{9}$.

EXAMPLE 4.108

Translate and solve: The sum of three-eighths and x is three and one-half.

Solution

Translate.

$$\underbrace{\frac{3}{8} + x}_{\text{The sum of three-eighths and } x} \quad \underbrace{=}_{\text{is}} \quad \underbrace{3\frac{1}{2}}_{\text{three and one-half}}$$

Use the Subtraction Property of Equality to subtract $\frac{3}{8}$ from both sides.	$\frac{3}{8} + x - \frac{3}{8} = 3\frac{1}{2} - \frac{3}{8}$
Combine like terms on the left side.	$x = 3\frac{1}{2} - \frac{3}{8}$
Convert mixed number to improper fraction.	$x = \frac{7}{2} - \frac{3}{8}$
Convert to equivalent fractions with LCD of 8.	$x = \frac{28}{8} - \frac{3}{8}$
Subtract.	$x = \frac{25}{8}$
Write as a mixed number.	$x = 3\frac{1}{8}$

We write the answer as a mixed number because the original problem used a mixed number.

Check:

Is the sum of three-eighths and $3\frac{1}{8}$ equal to three and one-half?

	$\frac{3}{8}+3\frac{1}{8}\stackrel{?}{=}3\frac{1}{2}$
Add.	$3\frac{4}{8}\stackrel{?}{=}3\frac{1}{2}$
Simplify.	$3\frac{1}{2}=3\frac{1}{2}$ ✓

The solution checks.

TRY IT 4.215 Translate and solve: The sum of five-eighths and x is one-fourth.

TRY IT 4.216 Translate and solve: The difference of one-and-three-fourths and x is five-sixths.

MEDIA

ACCESS ADDITIONAL ONLINE RESOURCES

Solve One Step Equations With Fractions (https://www.openstax.org/l/24SolveOneStep)

Solve One Step Equations With Fractions by Adding or Subtracting (https://www.openstax.org/l/24OneStepAdd)

Solve One Step Equations With Fraction by Multiplying (https://www.openstax.org/l/24OneStepMulti)

SECTION 4.7 EXERCISES

Practice Makes Perfect

Determine Whether a Fraction is a Solution of an Equation

In the following exercises, determine whether each number is a solution of the given equation.

498. $x-\frac{2}{5}=\frac{1}{10}$:
(a) $x=1$ (b) $x=\frac{1}{2}$
(c) $x=-\frac{1}{2}$

499. $y-\frac{1}{3}=\frac{5}{12}$:
(a) $y=1$ (b) $y=\frac{3}{4}$
(c) $y=-\frac{3}{4}$

500. $h+\frac{3}{4}=\frac{2}{5}$:
(a) $h=1$ (b) $h=\frac{7}{20}$
(c) $h=-\frac{7}{20}$

501. $k+\frac{2}{5}=\frac{5}{6}$:
(a) $k=1$ (b) $k=\frac{13}{30}$
(c) $k=-\frac{13}{30}$

Solve Equations with Fractions using the Addition, Subtraction, and Division Properties of Equality

In the following exercises, solve.

502. $y+\frac{1}{3}=\frac{4}{3}$

503. $m+\frac{3}{8}=\frac{7}{8}$

504. $f+\frac{9}{10}=\frac{2}{5}$

505. $h+\frac{5}{6}=\frac{1}{6}$

506. $a-\frac{5}{8}=-\frac{7}{8}$

507. $c-\frac{1}{4}=-\frac{5}{4}$

508. $x-\left(-\frac{3}{20}\right)=-\frac{11}{20}$

509. $z-\left(-\frac{5}{12}\right)=-\frac{7}{12}$

510. $n-\frac{1}{6}=\frac{3}{4}$

511. $p-\frac{3}{10}=\frac{5}{8}$

512. $s+\left(-\frac{1}{2}\right)=-\frac{8}{9}$

513. $k+\left(-\frac{1}{3}\right)=-\frac{4}{5}$

514. $5j=17$

515. $7k=18$

516. $-4w=26$

517. $-9v=33$

Solve Equations with Fractions Using the Multiplication Property of Equality

In the following exercises, solve.

518. $\frac{f}{4} = -20$

519. $\frac{b}{3} = -9$

520. $\frac{y}{7} = -21$

521. $\frac{x}{8} = -32$

522. $\frac{p}{-5} = -40$

523. $\frac{q}{-4} = -40$

524. $\frac{r}{-12} = -6$

525. $\frac{s}{-15} = -3$

526. $-x = 23$

527. $-y = 42$

528. $-h = -\frac{5}{12}$

529. $-k = -\frac{17}{20}$

530. $\frac{4}{5}n = 20$

531. $\frac{3}{10}p = 30$

532. $\frac{3}{8}q = -48$

533. $\frac{5}{2}m = -40$

534. $-\frac{2}{9}a = 16$

535. $-\frac{3}{7}b = 9$

536. $-\frac{6}{11}u = -24$

537. $-\frac{5}{12}v = -15$

Mixed Practice

In the following exercises, solve.

538. $3x = 0$

539. $8y = 0$

540. $4f = \frac{4}{5}$

541. $7g = \frac{7}{9}$

542. $p + \frac{2}{3} = \frac{1}{12}$

543. $q + \frac{5}{6} = \frac{1}{12}$

544. $\frac{7}{8}m = \frac{1}{10}$

545. $\frac{1}{4}n = \frac{7}{10}$

546. $-\frac{2}{5} = x + \frac{3}{4}$

547. $-\frac{2}{3} = y + \frac{3}{8}$

548. $\frac{11}{20} = -f$

549. $\frac{8}{15} = -d$

Translate Sentences to Equations and Solve

In the following exercises, translate to an algebraic equation and solve.

550. n divided by eight is -16.

551. n divided by six is -24.

552. m divided by -9 is -7.

553. m divided by -7 is -8.

554. The quotient of f and -3 is -18.

555. The quotient of f and -4 is -20.

556. The quotient of g and twelve is 8.

557. The quotient of g and nine is 14.

558. Three-fourths of q is 12.

559. Two-fifths of q is 20.

560. Seven-tenths of p is -63.

561. Four-ninths of p is -28.

562. m divided by 4 equals negative 6.

563. The quotient of h and 2 is 43.

564. Three-fourths of z is 15.

565. The quotient of a and $\frac{2}{3}$ is $\frac{3}{4}$.

566. The sum of five-sixths and x is $\frac{1}{2}$.

567. The sum of three-fourths and x is $\frac{1}{8}$.

568. The difference of y and one-fourth is $-\frac{1}{8}$.

569. The difference of y and one-third is $-\frac{1}{6}$.

Everyday Math

570. **Shopping** Teresa bought a pair of shoes on sale for $\$48$. The sale price was $\frac{2}{3}$ of the regular price. Find the regular price of the shoes by solving the equation $\frac{2}{3}p = 48$

571. **Playhouse** The table in a child's playhouse is $\frac{3}{5}$ of an adult-size table. The playhouse table is 18 inches high. Find the height of an adult-size table by solving the equation $\frac{3}{5}h = 18$.

Writing Exercises

572. Example 4.100 describes three methods to solve the equation $-y = 15$. Which method do you prefer? Why?

573. Richard thinks the solution to the equation $\frac{3}{4}x = 24$ is 16. Explain why Richard is wrong.

Self Check

ⓐ *After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.*

I can...	Confidently	With some help	No-I don't get it!
determine whether a fraction is a solution of an equation.			
solve equations with fractions using the addition, subtraction, and division properties of equality.			
solve equations using the multiplication property of equality.			
translate sentences to equations and solve.			

ⓑ *Overall, after looking at the checklist, do you think you are well-prepared for the next Chapter? Why or why not?*